MULTILINGUAL SOCIAL NETWORK ANALYSIS PIPELINE

An End-to-End Data & Visual Analytics Project

Subject: Data and Visual Analytics Using AI

Subject Code: 22CAE03

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Class: AIML-J 6th Semester

Pipeline Overview:

1. Initial Setup & GPU
2. Data Generation (500K Records)
3. Data Loading & Inspection
4. Exploratory Data Analysis
5. Data Preprocessing
6. Text Analysis
7. Visualization
8. Feature Engineering
9. Dimensionality Reduction
10. Classification Models
11. Evaluation Metrics
12. Prediction
13. Network Graph Analysis
14. Results & Insights

Aim & Description:

Aim

To design and implement a complete data analytics pipeline for multilingual social media data.

Project Description

- Analyze user-generated content from social networks
- Perform NLP on multilingual captions
- Study engagement patterns (likes, comments, shares)
- Build graph-based user relationships

Tools Used

- Python, Pandas, NumPy
- Scikit-learn
- Transformers (BERT)
- NetworkX
- Matplotlib, Seaborn

Step 1 – Initial Setup & GPU

Environment Setup

- Google Colab used for execution
- Installed required libraries

GPU Configuration

- CUDA Enabled: Yes
- Device: Tesla T4

Libraries Installed

- transformers, torch
- indic-nlp, networkx
- matplotlib, seaborn

Sample Code

```
import torch
print('CUDA Available:',
torch.cuda.is_available())
if torch.cuda.is_available():
  print('Device Name:',
torch.cuda.get_device_name(0))

# Install libraries
import subprocess
subprocess.check_call([sys.executable,
'-m','pip','install','-q','transformers'])
```

Steps 2–3: Data Generation & Loading

Data Generation

- Generated 500,000 synthetic records
- Multilingual dataset (English + Indic languages)

Dataset Format

- Stored in JSON format

Data Components

- User Profile
- Content Data
- Interaction Data
- Network Relationships
- Engagement Metrics

Data Loading

- Loaded JSON dataset
- Verified structure using sample records

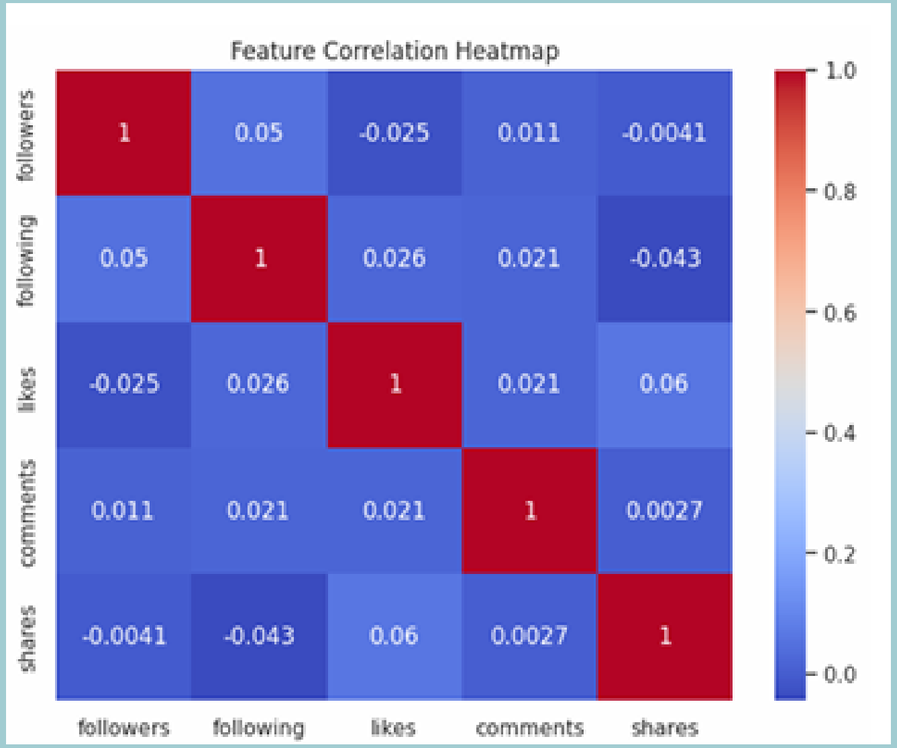
Step 4 – Exploratory Data Analysis

EDA Performed

- Language distribution
- Caption length analysis
- Engagement metrics distribution
- Followers vs engagement

Key Observations

- No missing values
- Balanced multilingual dataset
- Engagement values randomly distributed



Step 5 – Data Preprocessing

Text Cleaning Steps

1. Removed URLs
2. Removed hashtags & mentions
3. Removed special characters
4. Cleaned text for modeling

Example

Original:

“आज कड़ी मेहनत कर रहा हूँ #hi”

Cleaned:

“आज कड महनत कर रह ह”

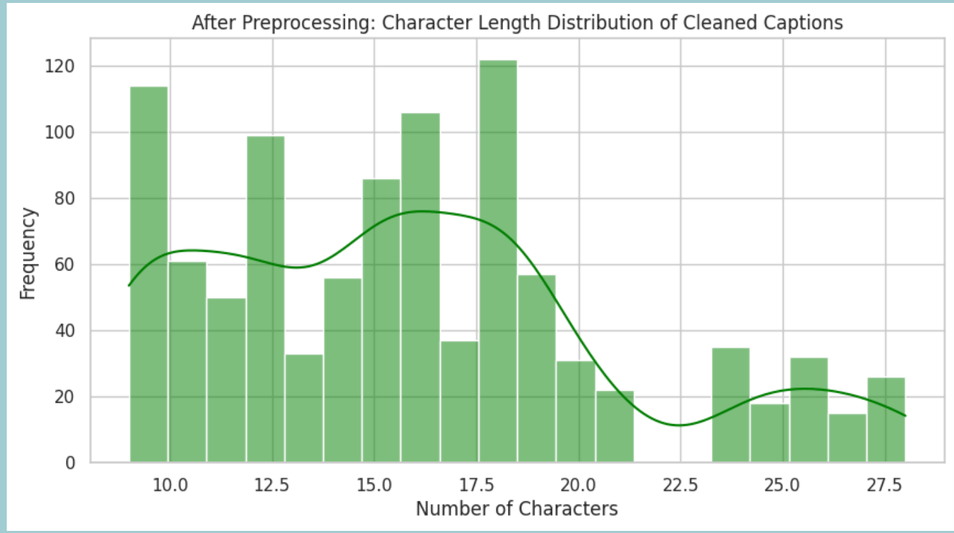
Step 6 – Visualization

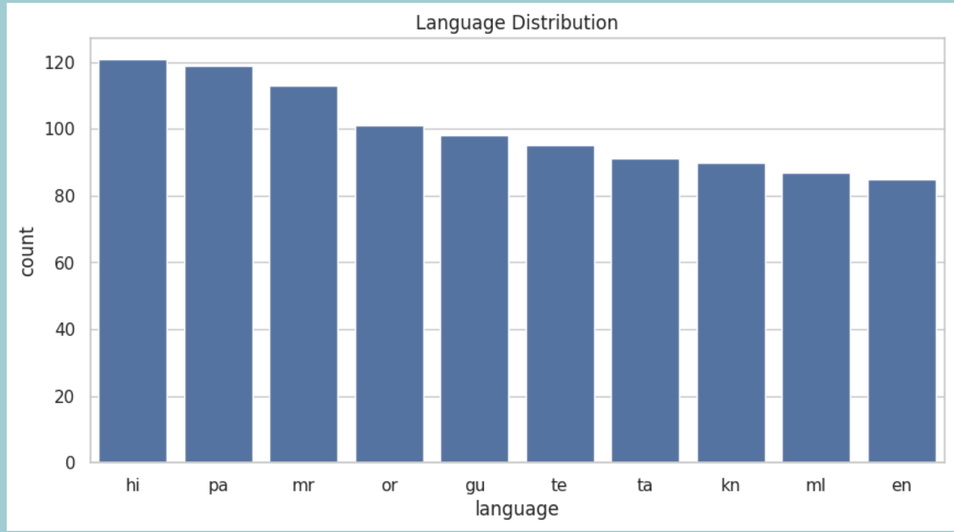
Visualizations Used

- Language distribution (count plot)
- Likes & comments histogram
- Correlation heatmap
- Caption length distribution

Key Insight

- Engagement metrics show no strong correlation
- Confirms synthetic random dataset





Step 7 – Feature Engineering

Techniques Used

TF-IDF

- Captures word importance

Bag of Words

- Frequency-based representation

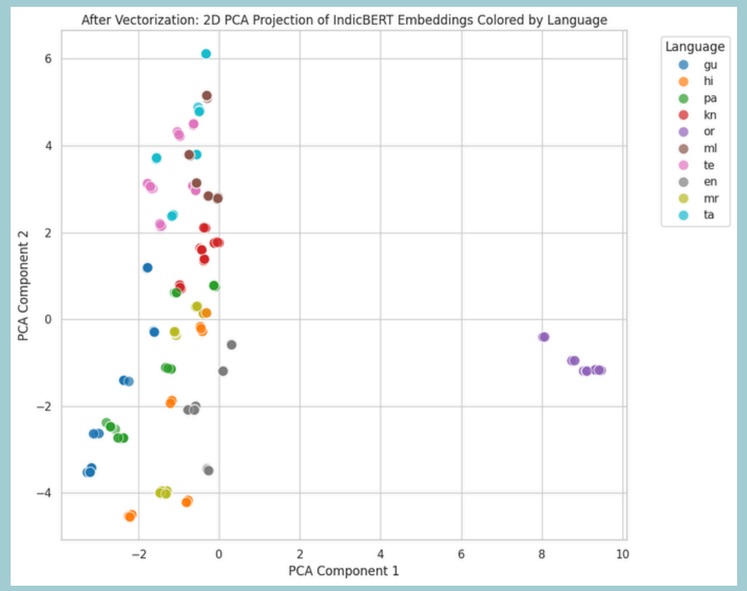
BERT Embeddings

- Captures semantic meaning
- Multilingual understanding

Feature Sizes

- TF-IDF: (500, 160)
- BoW: (500, 160)
- BERT: (500, 768)

Step 8 – Dimensionality Reduction



Methods Used

PCA

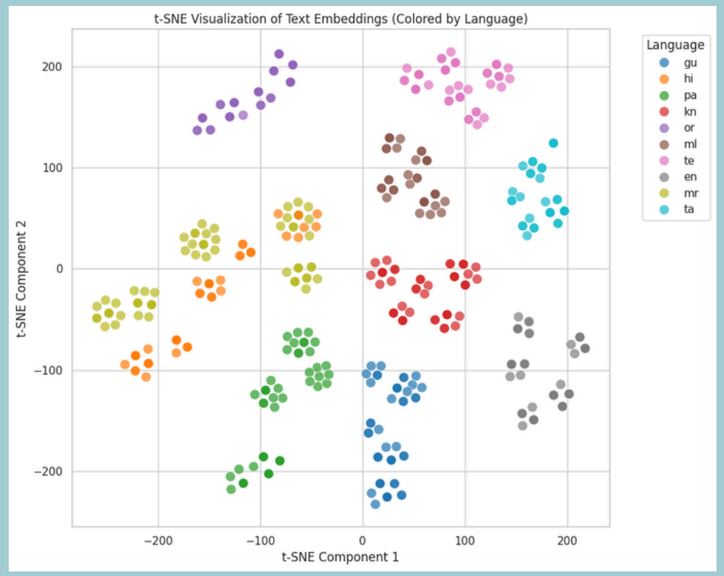
- Reduced dimensions to 2D

t-SNE

- Visualized language clusters

Outcome

- Clear grouping of languages
- Effective visualization of embeddings



Step 9 – Classification Models

Machine Learning Models

- Logistic Regression
- Naive Bayes
- Random Forest

Deep Learning Model

- Artificial Neural Network (ANN)

Task

- Language classification of captions

Step 10 – Evaluation Metrics

Performance Results

Machine Learning Models

- Accuracy ≈ 96.5%

Deep Learning Model

- Accuracy ≈ 97.5%

Metrics Used

- Accuracy
- Precision
- Recall
- F1-score

- Successfully processed 500K records
- Multilingual NLP pipeline implemented
- High classification accuracy (~97%)
- Effective visualization using PCA & t-SNE
- Network graph revealed user relationships

Step 11 – Prediction

Test Input

"आज मौसम बहुत अच्छा है"

Predicted Output

Language: Hindi

Conclusion:Model successfully predicts unseen multilingual text

Step 12 – Network Graph Analysis

Graph Construction

- Built using NetworkX

Graph Components

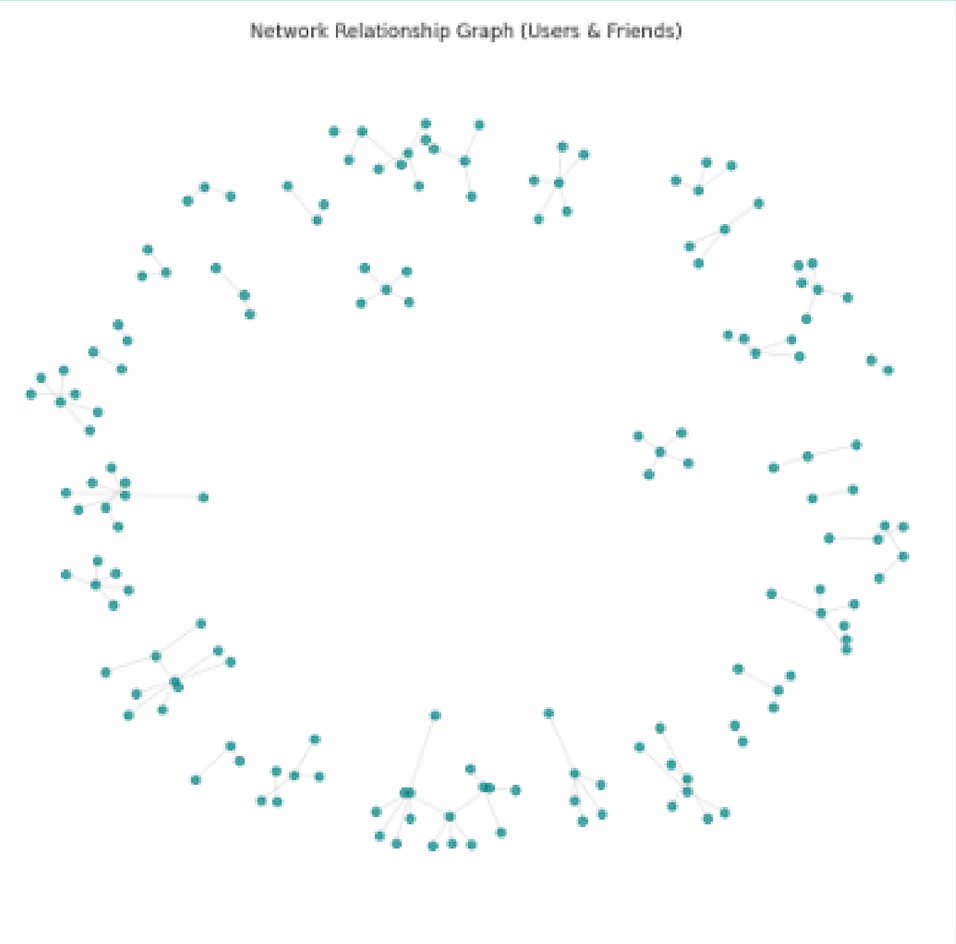
- Nodes → Users
- Edges → Connections

Analysis

- Degree centrality
- Influential users identified

Top Users

- Based on centrality score



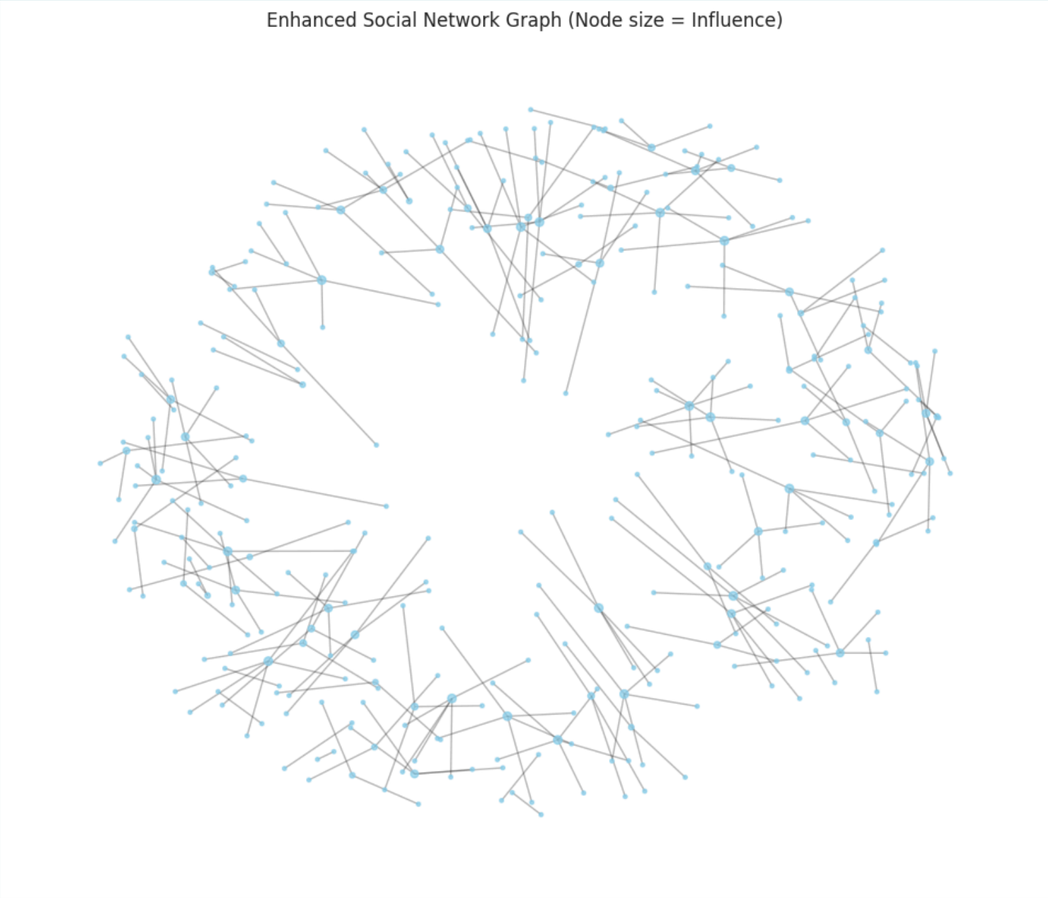
Step 13 – Visualization of Network

Graph Features

- Spring layout visualization
- Node size = Influence

Insights

- Sparse network structure
- Few highly connected users



Conclusion

- Built complete end-to-end data pipeline
- Combined NLP + ML + Graph Analysis
- Extracted meaningful insights from social data

Key Learning

- Multilingual data can be effectively analyzed using modern NLP techniques

THANK YOU